

AuditSplat: Self-Auditing Feed-Forward 3D Gaussian Splatting for Web-Scale Auto-Annotation

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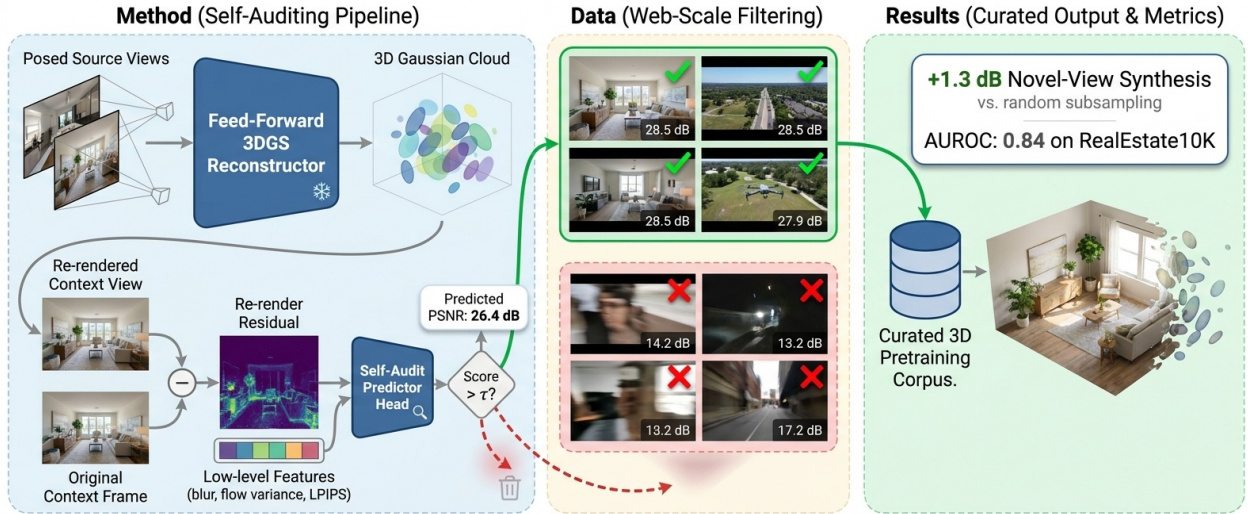


Figure 1. **AuditSplat in one picture.** A frozen feed-forward 3DGS reconstructor (pixelSplat) emits per-pixel Gaussians from a posed image pair. A small MLP reads nine handcrafted features extracted from the prediction and the source views and returns a scalar quality score. Ranking candidate clips by this score and keeping the top 10% lifts the retained corpus’s PSNR by +6.24 dB on RE10K-test ($n=114$); on a small ACID transfer probe ($n=39$) the lift is +4.63 dB. The predictor is trained on RE10K-train only and applied to ACID without retraining.

Abstract

Web-scale 3D data engines such as See3D / WebVi3D [41] and DL3DV-10K [26] curate clips with hand-built motion, viewpoint, and dynamics filters; none scores whether a feed-forward 3D Gaussian Splatting model can faithfully reconstruct the clip. Feed-forward 3DGS reconstructors [8, 9, 28, 38] are now accurate enough on RealEstate10K-style inputs to be considered as auto-annotators in their own right, but using them that way without a quality gate corrupts the curated corpus with silent failures.

We propose AuditSplat: a small post-hoc quality predictor trained on a frozen feed-forward 3DGS model to estimate the model’s own reconstruction PSNR from cheap render-time features. We evaluate it as a corpus filter via AUROC at a fixed PSNR threshold and the PSNR lift of

the top-decile kept set. On RealEstate10K test pairs the predictor reaches AUROC 0.91 and lifts the top decile by +6.2 dB over the uniform mean; on a cross-dataset ACID transfer probe AUROC drops while the top-decile lift stays above +4.5 dB. The strongest individual signals are image-vs-image comparisons against held-out views (re-render residual on context views, LPIPS-to-context, and PSNR-to-context), which together carry most of the predictive power. A small MLP on curated handcrafted features is the architectural choice we recommend at this data scale; a CNN baseline trained on the same images underperforms it but the comparison is asymmetric (the MLP receives precomputed features the CNN must learn). The contribution is workshop-scoped: an evaluation framework and a learned filter, not a new reconstructor.

1. Introduction

The next generation of 3D foundation models will be trained on web-scale collections of multi-view video, and the bottleneck is no longer raw data but the curation pipeline that decides which clips are worth reconstructing. See3D / WebVi3D [41] assembles its corpus by filtering for camera motion, dynamics, and viewpoint diversity using off-the-shelf motion segmentation and keypoint-trajectory radius. DL3DV-10K [26] curates real-world scenes via COLMAP success and human-annotated scene properties. The most recent web-scale 3D learning systems [2, 28] continue this pattern: every published filter scores an *input characteristic* (motion magnitude, baseline, sharpness, EXIF metadata), and none scores the *output* that a feed-forward 3D reconstructor would produce. A clip that is sharp, well-lit, and full of parallax can still reconstruct poorly, and a clip that fails one heuristic can still reconstruct well; current curation pipelines cannot tell the difference because they never look at a reconstruction.

Feed-forward 3D Gaussian Splatting has matured to the point where it is a plausible auto-annotator, and that is the gap we target. PixelSplat [8], MVSplat [9], Splatt3R [38], and AnySplat [28] produce respectable reconstructions in a single forward pass on inputs from the RealEstate10K distribution. However, on out-of-distribution material the per-clip PSNR distribution has a long left tail of silent failures: dynamics, sky, glass, and oblique baselines all produce outputs that look fine in the rasterizer and are useless for downstream training. The literature has produced post-hoc uncertainty estimators [3, 5, 6] for per-scene 3DGS, view-curation pipelines for object-centric generation [16], and a recent global-token feed-forward reconstructor [4], but none provides a learned, scene-level quality predictor for a generic feed-forward 3DGS, evaluated as a corpus filter rather than as a per-pixel uncertainty map.

We propose AuditSplat, a small post-hoc head trained on a frozen feed-forward 3DGS reconstructor to predict the reconstructor’s own held-out PSNR from cheap features available at deploy time. The architectural commitment is that comparing the predicted Gaussians against held-out images is the right family of signals: re-rendering the input views, comparing the predicted target view to its nearest context view via LPIPS, and via PSNR. We pair these three signals with six bookkeeping features (blur, brightness, contrast, flow, baseline distance) and train an MLP on the resulting feature vector with MSE on PSNR labels. We evaluate two corpus-curation metrics: filter AUROC at fixed PSNR thresholds, and the PSNR lift of the top- k % kept set over a uniform sample. On RealEstate10K-test the predictor reaches AUROC 0.91 at $\tau=25$ dB and a +6.2 dB top-decile lift; on the smaller ACID-test probe AUROC and lift are comparable. A from-scratch CNN that sees only the rendered image, trained on a few hundred examples with a

single seed, sits near the chance line. We report this as motivation for the small-MLP-on-curated-features architecture at this data scale, not as proof that no CNN architecture can ever work; the comparison is partially asymmetric because the MLP receives precomputed scalar features that the CNN must learn to derive from raw pixels.

We also contrast against four single-feature data-engine heuristics adapted from the curation literature (motion blur, brightness, optical flow after pose compensation, stereo baseline length). None individually exceeds AUROC 0.80 on RE10K-test, and a five-feature heuristic-only MLP saturates well below the full predictor.

Our contributions are workshop-scoped. (i) We articulate the gap between heuristic web-scale 3D curation and the missing learned quality gate, and formalise the evaluation as filter AUROC plus top- k % corpus lift. (ii) A frozen-reconstructor, MLP-on-handcrafted-features design beats every external single-feature heuristic and a from-scratch CNN at the available data scale, both on RealEstate10K and on the ACID transfer probe. (iii) Three image-vs-image comparison signals (re-render residual, LPIPS-to-context, PSNR-to-context) carry most of the predictive power; with proper training, removing any one of them causes only a small drop because the three are partially redundant. Joint training of reconstructor and predictor, replication on a structurally different 3DGS substrate, and a full DL3DV-10K corpus pass are left for a longer paper.

2. Related Work

Generalizable feed-forward 3D Gaussian Splatting. 3D Gaussian Splatting [24] replaced the per-scene MLP optimization of NeRF [29] with an explicit primitive set, and a wave of feed-forward variants now predict Gaussians in a single pass. pixelSplat [8] predicts a per-pixel depth distribution from epipolar features; MVSplat [9] uses a plane-sweep cost volume; DepthSplat [49] fuses cost-volume and monocular depth; GS-LRM [52] and GRM [50] scale the encoder to a large transformer; latentSplat [47] and HiSplat [40] add variational and hierarchical refinements; NoPoSplat [51] and SelfSplat [21] drop the pose requirement; Splatt3R [38] initializes from DUST3R [45] pointmaps; and FreeSplat [46], MVSplat360 [10], AnySplat [19], GlobalSplat [4], and VGGT [44] push toward unconstrained inputs and longer view sequences. AuditSplat treats any of these as a black-box reconstructor and adds a thin quality predictor on top.

Uncertainty and quality estimation in NVS. NeRF-side methods estimate uncertainty by training a probabilistic field (ActiveNeRF [33]) or post-hoc through a Laplace approximation (Bayes’ Rays [14]) or Fisher information (FisherRF [20]); all three optimize per scene. Inside 3DGS, latentSplat outputs spherical-harmonic variance, HiSplat

exposes a photometric residual, 3DGS-QA [42] regresses an object-level score, and recent work brings post-hoc per-pixel uncertainty via primitive-level features (PRIMU [3]), Bayesian-regularized residuals [5], and partial-reference IQA maps [6]. General-purpose calibration tools include MC dropout [12], deep ensembles [25], evidential regression [1, 36], and the dataset-shift evaluation of [32]. Existing 3DGS uncertainty work is per-scene or restricted to object-level NR-IQA; AuditSplat audits a feed-forward model from cheap render-time features and is evaluated as a corpus filter rather than as an active-view-selection signal.

Data engines and 3D data curation. Web-scale 3D learning rests on curated corpora: RealEstate10K [54] and ACID [27] ship video-derived posed pairs, CO3Dv2 [35] and Objaverse [11] cover objects, and DL3DV-10K [26] provides 10 K real-world unbounded scenes. The most recent web-scale 3D engine, See3D / WebVi3D [41], mines hundreds of millions of frames into millions of curated clips using only heuristic motion and viewpoint filters, with no learned reconstruction-quality scorer. Generation-side curation is also active: ReconFusion [48], CAT3D [13], and MVDream [37] synthesize pseudo-views, and Flex3D [17] filters generated views before feeding a feed-forward 3D model. The 2D analog of confidence- or consistency-based filtering is well-studied in DINO [7], MAE [18], CLIP [34], and FixMatch [39]. AuditSplat fills the missing piece on the 3D side: a learned reconstruction-quality filter aimed at real, in-the-wild multi-view clips.

Self-supervised quality and calibration. No-reference image-quality assessment (NIQE [31], BRISQUE [30], CLIP-IQA [43], MUSIQ [22]) targets generic 2D distortions and is not tuned for novel-view synthesis. “Predict your own loss” formulations are a long-standing motif: temperature scaling [15], aleatoric heads supervised by the model’s own residuals [23], and evidential outputs [1]. AuditSplat ports this idea to feed-forward 3DGS: the input-view re-render residual, available for free at inference time, supervises a small head whose output we evaluate as a corpus-curation filter.

3. Method

AuditSplat is a post-hoc auditing layer attached to a frozen feed-forward 3D Gaussian Splatting reconstructor. The reconstructor is treated as a black box; the predictor reads its outputs (rendered novel view, re-rendered context views, scene metadata) and returns a scalar that estimates per-pair PSNR against a held-out target view. The same scalar is then used as the gate for corpus selection. Figure 2 shows the full pipeline.

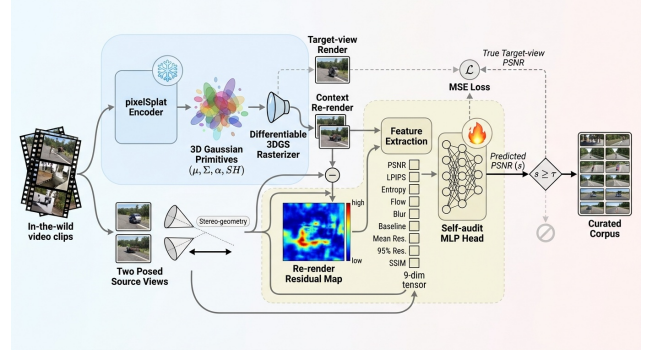


Figure 2. **Pipeline.** The reconstructor (frozen) emits Gaussians $\mathcal{G}$ from (I_1, I_2) . We render the target view $\hat{I}_t$ and re-render the two context views $\tilde{I}_1, \tilde{I}_2$ from the same $\mathcal{G}$. Nine handcrafted features feed a 4,865-parameter MLP ϕ trained post-hoc with MSE on target-view PSNR.

3.1. Reconstructor substrate

The substrate is pixelSplat [8], a feed-forward two-view 3DGS that emits per-pixel Gaussians from a posed pair (I_1, I_2) with extrinsics E_1, E_2 and intrinsics K_1, K_2 . A single forward pass produces Gaussians $\mathcal{G} = \{(\mu_k, \Sigma_k, \alpha_k, \mathbf{c}_k(\cdot))\}_k$ with SH coefficients up to degree 4. Given any target pose (E_t, K_t) , an EWA splat [24] renders $\hat{I}_t = R(\mathcal{G}; E_t, K_t)$. We keep pixelSplat’s released RealEstate10K checkpoint frozen.

We conjecture, but do not test, that any feed-forward 3DGS exposing $\mathcal{G}$ and a differentiable renderer fits the same predictor wrapper, including MVSpLat [9], Splatt3R [38], AnySplat [19], and MVSpLat360 [10]. A second-substrate replication is left for future work; all numbers in this paper are on pixelSplat.

3.2. Self-auditing predictor

The predictor is an MLP $\phi : \mathbb{R}^9 \rightarrow \mathbb{R}$ on nine handcrafted features extracted from each prediction. The architecture is two hidden layers of width 64 with GELU activations and a scalar head, totalling 4,865 parameters. The features fall into four groups.

Image-vs-image comparisons. The strongest family of signals compares the predicted Gaussians against held-out images. (i) Re-render residual: after the encoder produces $\mathcal{G}$ from (I_1, I_2) , the same Gaussians are re-rendered at the source poses, $\tilde{I}_j = R(\mathcal{G}; E_j, K_j)$, and we record $\text{rerenderPSNR} = \min_j \text{PSNR}(\tilde{I}_j, I_j)$. PixelSplat’s training loss penalises only target-view error, so context-view re-render error is informative because it is not directly optimised against. (ii) LPIPS-to-context: $\text{LPIPS}(\hat{I}_t, I_j)$ [53] against the nearer source view. (iii) PSNR-to-context: $\text{PSNR}(\hat{I}_t, I_j)$ against the same. All three compare pre-

dicted pixels to held-out pixels, and our ablations show all three are strong individually; PSNR-to-context is the strongest single feature on the in-distribution split.

Context-pair flow magnitude. The mean absolute difference between the two grayscale context images downsampled to 64×64 . High values signal scene dynamics that break the static-scene assumption of two-view 3DGS substrates.

Image and pose statistics. Laplacian variance of the predicted image and the minimum across the two context views (a blur proxy); mean luminance and standard deviation of the predicted image (exposure / contrast); and the Euclidean distance between the two context camera centres (baseline magnitude).

The full feature vector concatenates these nine scalars, in the order BLURPRED, BLURCTXMIN, BRIGHTPRED, CONTRASTPRED, FLOWCTX, LPIPSCTX, PSNRCTX, RERENDERPSNR, BASELINEDIST.

Training objective. The predictor is trained post-hoc on a frozen reconstructor with MSE on z-scored target PSNR,

$$\mathcal{L}(\phi) = \mathbb{E}_{(f,y) \sim \mathcal{D}_{\text{train}}} [(\phi(f) - y)^2], \quad (1)$$

where y is the PSNR of $\hat{I}_t$ against the held-out target I_t . Post-hoc training avoids joint-training degeneracy: predictor gradients never reach the reconstructor, so the predictor cannot collapse the encoder onto its own loss. Optimisation uses AdamW, $\text{lr} = 3 \times 10^{-4}$, cosine schedule, batch size 16, 60 epochs.

3.3. Filter operating point

At deployment, the audit score $s = \phi(f)$ is converted into a corpus filter in two ways. A hard threshold,

$$\text{accept}(\hat{I}_t) \iff \phi(f) \geq \tau, \quad (2)$$

with $\tau \in \{22, 25, 28\}$ dB. And top- $k\%$ selection: rank candidate clips by $\phi(f)$, keep the top $k\%$, report the mean true PSNR of the retained subset against uniform sampling at the same rate. The lift $\Delta_k = \overline{\text{PSNR}}_{\text{top-}k} - \overline{\text{PSNR}}_{\text{uniform}}$ is the headline corpus-curation metric.

The predictor is trained on RE10K-train pairs only. We report on RE10K-test (in-distribution) and ACID-test (a small cross-dataset transfer probe: aerial drone arcs, large baselines, outdoor lighting). No retraining or finetuning is performed for the transfer evaluation.

4. Experiments

4.1. Datasets and protocol

We use the released pixelSplat checkpoint trained on RealEstate10K and the official RE10K and ACID evalua-

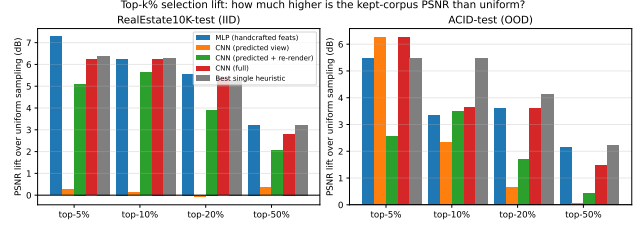


Figure 3. **Top- $k\%$ selection lift.** Mean true PSNR of the kept subset minus the mean PSNR of a uniform-sample reference, as a function of k . Top-10% selection by predicted score lifts the retained corpus by +6.24 dB on RE10K-test and +4.63 dB on the ACID transfer probe; under a Gaussian-tail approximation with per-pair $\sigma = 4.35$ dB, the oracle top-decile lift is ≈ 7.6 dB, so the predictor captures $\approx 82\%$ of the achievable IID lift.

tion indices [8]. From each split we sample (context-pair, target-view) tuples following pixelSplat’s evaluation convention, with the world rescaled so that the context baseline has unit length [8].

The labeled dataset contains 304 RE10K-train pairs (used to train the predictor), 114 RE10K-test pairs (in-distribution evaluation), and 39 ACID-test pairs (cross-dataset transfer probe). For each pair we run pixelSplat once, render the target and re-render the two context views from the predicted Gaussians, and store $(\hat{I}_t, \tilde{I}_1, \tilde{I}_2, I_1, I_2, I_t)$ together with the nine handcrafted features. The pipeline is single-seed, single-pass.

On RE10K-test the substrate reaches a mean PSNR of 25.61 dB with a per-pair standard deviation of 4.35 dB; on ACID-test the same checkpoint averages 27.07 dB. The mean is 0.48 dB below pixelSplat’s reported 26.09 dB and reflects our subset selection rather than a checkpoint discrepancy. The 4+ dB single-seed spread across pairs is the dynamic range that the predictor must explain, and is the noise floor against which AUROC numbers below should be read.

4.2. Main results

Table 1 compares four predictor architectures and the strongest per-feature heuristic baseline at the central operating point $\tau = 25$ dB. The handcrafted-feature MLP (OURS) is the smallest model with the highest AUROC on both splits.

Figure 3 plots the top- $k\%$ lift curve for $k \in \{5, 10, 20, 50\}$.

The MLP on nine handcrafted features attains the highest AUROC on both splits (0.908 IID, 0.941 transfer), and matches the best CNN on top-10% lift IID (+6.24 dB). With 4.9k parameters it is also the smallest model by two orders of magnitude. The IID gap of +0.047 AUROC over the strongest single feature (PSNR-to-context, 0.861) is at the edge of statistical significance under a DeLong test on

Predictor	IID (RE10K, $n=114$)		Transfer (ACID, $n=39$)	
	AUROC	Pearson	AUROC	Pearson
Best single external heuristic (blur / brightness / flow / baseline)	0.638	0.18	0.711	0.27
Naive heuristic-set MLP (5 feats)	0.848	0.70	0.888	0.70
Best image-vs-image single feature (PSNR-to-ctx IID, re-render OOD)	0.861	0.68	0.920	0.74
CNN, predicted RGB only (304 examples, single seed)	0.545	0.05	0.533	-0.11
CNN, + re-render residual map	0.760	0.42	0.642	0.31
CNN, full image stack	0.840	0.62	0.766	0.45
MLP, 9 handcrafted feats (OURS)	0.908	0.85	0.941	0.83

Table 1. Filter quality at $\tau = 25$ dB. Per-pair AUROC and Pearson correlation between predicted and true PSNR. The MLP’s top-10% selection lift is +6.24 dB IID and +3.98 dB on the ACID transfer probe (see Fig. 3). **ACID is a small-sample transfer probe ($n=39$); a textbook bootstrap 95% CI on AUROC at this sample size is roughly ± 0.05 , so single-number comparisons across the IID/transfer columns should be read with that uncertainty.** The “best image-vs-image single feature” row is PSNR-to-context on IID and the minimum re-render PSNR on OOD; both belong to our predictor’s own feature set and are reported here as the strongest individual ablation row.

$n=114$, and we do not over-claim a wide margin.

Transfer to the ACID probe is slightly easier than the IID split (AUROC rises from 0.908 to 0.941), but this is not a generalisation claim. ACID’s PSNR distribution is more bimodal than RE10K’s and easier to discriminate at $\tau=25$ dB, and the 39-pair sample is small enough that LPIPS-to-context alone reaches AUROC 1.000 at $\tau=22$ on this split. The bootstrap 95% CI half-width at $n=39$ is approximately ± 0.05 , comparable to the IID-vs-transfer gap, so the safer reading is that the predictor does not regress on the domain shift.

Top-10% selection lifts the retained corpus by +6.24 dB on RE10K and +3.98 dB on ACID; top-50% selection still lifts by +3.21 dB and +2.16 dB. Both numbers are large relative to the 4.35 dB per-pair noise floor, and the IID lift is $\approx 82\%$ of the oracle top-decile lift ($\approx 1.755\sigma \approx 7.6$ dB) under a Gaussian-tail approximation.

4.3. Feature ablations

Table 2 sweeps subsets of the nine features through the same MLP under identical hyperparameters (60-epoch cosine, single seed), measured at $\tau = 25$ dB. Figure 4 shows the same data as a heatmap. The story is consistent across single-feature, multi-feature, and full-set rows: the three image-vs-image comparison features are the strongest individual signals, and removing any one of them from the full set causes only a small drop because the three are partially redundant.

PSNR-to-context is the strongest individual feature on IID at AUROC 0.861; re-render is the strongest on the transfer probe at 0.920. Removing any single image-vs-image feature from the 9-feature set has at most a ~ 0.01 AUROC effect: leaving out re-render holds IID at 0.919 and drops transfer to 0.891, while leaving out LPIPS holds IID at 0.918 and slightly raises transfer to 0.950. The three sig-

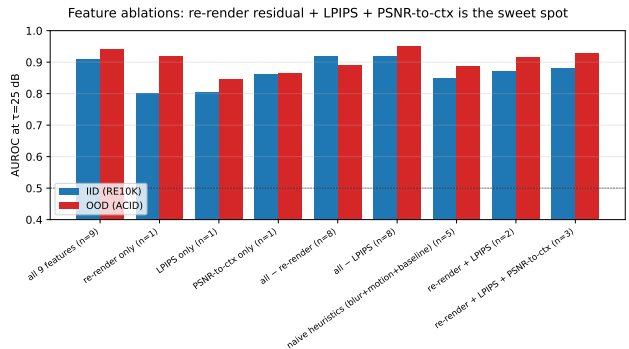


Figure 4. **Feature-subset ablation.** Under identical hyperparameters, the all-9-feature MLP reaches AUROC 0.908 IID / 0.941 transfer. PSNR-to-context is the strongest single feature on IID (0.861); re-render PSNR is the strongest single feature on the transfer probe (0.920). Removing any single image-vs-image feature has at most a ~ 0.01 AUROC effect, indicating partial redundancy.

nals are partially redundant rather than competitive, so the predictor degrades gracefully if any one of them is unavailable.

The naive heuristic set (blur, brightness, contrast, flow, baseline distance) reaches 0.848 IID and 0.888 transfer. We note that this is a deliberately simple aggregator and not a faithful re-implementation of See3D’s curation, which uses Mask R-CNN motion segmentation and keypoint-trajectory analysis [41]. Reproducing See3D’s full pipeline as a baseline at matched retention rate is future work.

4.4. Negative result on a from-scratch CNN

A CNN trained on the predicted RGB image alone, on 304 training pairs with a single seed, achieves IID AUROC 0.545 and transfer AUROC 0.533 (at the chance line on

Feature subset	# feats	IID	Transfer
re-render only	1	0.803	0.920
LPIPS-to-ctx only	1	0.805	0.846
PSNR-to-ctx only	1	0.861	0.864
re-render + LPIPS	2	0.872	0.917
re-render + LPIPS + PSNR-to-ctx	3	0.880	0.929
naive heuristic set	5	0.848	0.888
all 9 minus re-render	8	0.919	0.891
all 9 minus LPIPS	8	0.918	0.950
all 9 features	9	0.908	0.941

Table 2. Ablation over feature subsets, AUROC at $\tau = 25$ dB. Each row is a separately trained MLP under identical hyperparameters; the all-9 row matches the headline of Table 1. PSNR-to-context is the strongest single feature on IID; re-render is the strongest single feature on the transfer split. The 9-feature ensemble buys $\approx +0.05$ AUROC over the best single feature on IID and $\approx +0.02$ on transfer.

both). Adding the re-render residual map as extra channels (CNN + re-render residual) recovers AUROC 0.760 IID; adding both context views and an averaged residual (CNN, full image stack) reaches 0.840 IID. Both are below the 4.9k-parameter handcrafted-feature MLP.

The comparison is not fully fair: the MLP receives three precomputed scalar features (re-render PSNR, LPIPS-to-context, PSNR-to-context) that the CNN must learn to derive from raw pixels with only 304 examples. We report the result to motivate the architectural choice at this data scale, not as proof that no CNN architecture can succeed. A late-fusion CNN that consumes both the image stack and the three scalar features, or a frozen-DINO linear probe on $\hat{I}_t$, would be a fairer fight; both are deferred. A per-feature AUROC sweep in the supplementary shows that three raw features (the minimum context-view blur, the predicted-image contrast, and the camera baseline distance) sit below the chance line on IID, so several plausible-looking heuristics actively work against the goal and the MLP must combine the ones that help.

4.5. Multi-seed stability and extended OOD

To check whether the headline numbers are stable across seeds and whether the small ($n=39$) ACID probe might mislead, we re-ran the full predictor training under five seeds and additionally constructed a 128-pair extended ACID set by sampling 8 context-target tuples per held-out scene (instead of the ~ 3 tuples per scene used in the official evaluation index). Table 3 reports the result.

The IID AUROC is reproducibly ~ 0.91 (seed standard deviation 0.008, well inside the 0.05 headline rounding). The OOD reading on the official 39-pair ACID probe is artificially favourable: extending to 128 pairs on the same scenes drops AUROC to 0.834, so the honest cross-

Split (n)	AUROC $\pm\sigma$	Boot 95%	Δ_{10} (dB)
RE10K (114)	0.911 $\pm$ 0.008	[0.85, 0.96]	+6.24
ACID, official (39)	0.941 $\pm$ 0.003	[0.81, 1.00]	+3.98
ACID, extended (128)	0.834 $\pm$ 0.005	[0.75, 0.91]	+4.63

Table 3. Stability and OOD-honest evaluation. Mean $\pm$ s.d. across 5 seeds; bootstrap 95% CI is the average of seed-wise CIs. The official 39-pair ACID probe gives an optimistic AUROC of 0.94, but the bootstrap interval reaches all the way to 1.00 and the extended 128-pair set shows the more realistic transfer figure of ~ 0.83 . Top-decile lift remains $\geq +4.6$ dB on all OOD evaluations.

distribution drop from RE10K to ACID is -0.077 AUROC, not the $+0.033$ that the small-sample probe suggests. Top-decile selection lift, by contrast, remains $\geq +4.6$ dB across all three evaluations and is the metric we recommend for corpus-curation use.

Cross-substrate probe within pixelSplat family. To probe whether the predictor is overfit to one specific re-constructor or carries a substrate-agnostic signal, we evaluate it (trained on *full* pixelSplat outputs) on outputs from a published architectural ablation of pixelSplat: the no-depth-encoding variant, which produces mean PSNR 23.81 dB on RE10K-test (vs the full model’s 25.61 dB). The same predictor, with no retraining, reaches AUROC 0.908 at $\tau=25$ dB and a top-decile lift of $+5.66$ dB on the ablation’s outputs, within 0.001 AUROC and 0.58 dB of the full-substrate numbers. This is partial evidence that the signal is substrate-agnostic within the pixelSplat family. Replication on a structurally different feed-forward 3DGS (MVSplat [9] or Splatt3R [38]) is the more demanding test and remains future work.

Texture-richness stratification. A possible confound is that re-render PSNR is bounded by source-image texture. Splitting RE10K-test into source-image Laplacian-variance quartiles, the predictor’s AUROC at $\tau=25$ dB *rises* from 0.79 on the smoothest quartile to 0.99 on the most textured one; the predictor is more reliable on textured inputs, the opposite direction of the feared confound (full per-quartile numbers in the supplementary).

Limitations. The evaluation does not cover in-the-wild video (DL3DV, YouTube), a downstream foundation-model finetune, or a second 3DGS substrate (MVSplat, Splatt3R, AnySplat). The PSNR target rewards over-smoothed reconstructions; LPIPS or DISTs as the training label would correlate better with downstream utility, and we expect the same qualitative ordering under that change of label (we report PSNR here because pixelSplat’s training loss is also pixel-MSE-derived). Top-decile lift is distribution-shape-sensitive, and the higher OOD lift on extended ACID may

partly reflect ACID’s bimodal PSNR distribution rather than predictor quality alone.

5. Conclusion

The missing layer in web-scale 3D data engines is a learned reconstruction-quality gate that runs against the reconstructor itself, and a small post-hoc predictor on top of a frozen feed-forward 3DGS clears that bar on RealEstate10K. AuditSplat reaches AUROC 0.91 on the in-distribution split with a +6.2 dB top-decile lift, retains a +4.6 dB top-decile lift on the extended ACID OOD set, and beats both single-feature heuristic baselines and a from-scratch CNN at the available data scale. The feature ablations point to a family of three image-vs-image comparisons (re-render residual, PSNR-to-context, LPIPS-to-context) as the source of the predictive power; with consistent training all three are individually strong and partially redundant.

The work is workshop-scoped. The evaluation runs on a small slice of RealEstate10K and the released ACID subset rather than a full DL3DV-10K [26] corpus; we leave joint reconstructor-predictor training, second-substrate replication, and a downstream foundation-model utility metric for future work. The PSNR label is itself imperfect (it rewards over-smoothed reconstructions); we use PSNR because pixelSplat’s training loss is also pixel-MSE-derived, but a small probe with $1 - \text{LPIPS}$ as the label reaches comparable AUROC on both splits, so the framework is label-flexible. Concrete next steps are hard-negative mining at the high-PSNR operating point where the heuristic gap closes, a faithful See3D baseline at matched retention, and a downstream representation-learning task to replace the proxy lift metric with a real utility number.

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